

From Noise to Narrative:

A Visualization Tool for DDoS Events Across PEST dimension

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Abstract

DDoS attacks form a severe threat to organizations, as their services can be made unavailable at any moment, resulting in major direct and indirect damage. This research uses a multi output regression model on a DDoS Google Alert dataset, in order to validate event level insights in article level data. The research provides a LLM powered annotated dataset and uses a Support Vector Regression Model to validate. The annotations record severity of damaging impact on Political, Economical, Social, and Technological (PEST) dimensions. In this research an artifact that uses semantic embedding to cluster articles into an event level has been created and published. The regressor and LLM layer are included in the artifact to ensure valid results. Results show that LLM annotation yields low inter-rater agreement with human labels, and embedding-based regression fails to predict PEST scores reliably. Despite this, semantic clustering successfully identifies discrete DDoS events from noisy article collections, and the resulting visualization tool demonstrates the viability of automated, event-level impact analysis from open-source news data. The research can be used to compute the PEST severity of all DDoS events on the internet. This collection can be used to gain a better understanding of DDoS attacks.

ACM Reference Format:

Jesse van Dijk. 2026. From Noise to Narrative: A Visualization Tool for DDoS Events Across PEST dimension. In *Proceedings of the Twente Student Conference on IT (TScIT)*. ACM, New York, NY, USA, 9 pages.

Keywords: Machine Learning, Google Alerts, Regression, DDoS, Event Detection, Semantic Embedding

1 Introduction

In 2016, one of the most well-known and illustrative DDoS attacks happened; more than a million devices attacked DNS provider Dyn, resulting in widespread outages affecting major online services such as Netflix, Airbnb, and the BBC [1]. More recently, social media service Bluesky was attacked for 24 hours on April 15th 2026 by a pro-Iran hacktivist group[12]. Distributed Denial of Service attacks, or DDoS attacks are attempts to make digital infrastructure unavailable by flooding it with requests from multiple sources. This research aims to investigate how several dimensions effect the social ‘popularity’ of a DDoS event. This provides insight into public perception of DDoS related events, how this changes historically, and the

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Table 1: Comparison of related work

	PEST impact	Data-driven	Event-level view	ML algorithms	LLM annotation
Kamerman [11]	x	✓	x	✓	x
Agrafiotis et al. [2]	✓	x	✓	x	x
Gandhi et al. [8]	✓	✓	✓	x	x
Chen et al. [5]	✓	✓	x	x	x
Reese et al. [17]	x	✓	x	x	✓
This paper	✓	✓	✓	✓	✓

socioeconomic damage associated with these events. These insights may ultimately support more informed cybersecurity cost-benefit analyses and decision-making.

- RQ1: To what extent does news coverage of a DDoS event reflect its PEST-dimension impact?
- RQ2: How effectively can event-level groupings of DDoS events be constructed from article-level news data using semantic embedding and clustering?
- RQ3: How do we create a measurement tool to visualize the impact of an event based on the annotated dataset?

First, the relevance of the paper is established by identifying gaps in the existing literature. Next, the methodology is presented, including the machine learning pipeline used to transform articles into event-level data, as well as the regression and validation models applied to the annotated dataset. The data source, preprocessing techniques, and measured dimensions are then described. Finally, the results are presented, followed by the conclusions and their implications.

2 Related work

To the best of our knowledge, no prior work has investigated regression models on a Google Alert dataset for assessing the societal impact of DDoS events. Therefore, related work was selected based on methodological and conceptual similarities. The selected studies cover machine learning on Google Alert data, frameworks for assessing cyber harm, analyses of the motivations behind cyberattacks, studies on the relationship between different impact dimensions, and the use of large language models (LLMs) for annotation and evaluation.

- Kamerman [11]: Comparison of classification models on a Google Alert dataset, classifying attacks and arrests.
- Agrafiotis, et al. [2]: Defying impact categories of cyberharm and giving a taxonomical framework for businesses to use.
- Gandhi, et al. [8]: Explain political and social motivation behind cyberattacks.
- Chen, et al. [5]: Find the relationship of different impact dimensions on cyberattacks.
- Reese, et al. [17]: Comparing LLM judgment against human judgment.

Table 1 compares these studies with the proposed approach across the main characteristics considered in this work. While individual studies address one or more of these aspects, none combines PEST-based impact assessment, a data-driven event-level analysis, machine learning, and LLM-assisted annotation

3 Dataset annotation

The dataset used in this research, is a subsection of a database generated from Google Alerts messages. Google Alert is a service which notifies users whenever new content matching a specified keyword becomes available online. For this research, alerts containing the keywords "DDoS" and "Distributed Denial of Service" were collected. The complete database consists of 235.913 articles and stores the article ID, publication timestamp, URL, and the content provided by the Google Alerts. Of this database, a selection of 1000 recent articles have been selected for this research. A custom Python script was used to scrape the title and full text of each article, which were subsequently added to the dataset.

3.1 Large Language Model annotation

From this dataset, we want to gain knowledge of the impact of several impact dimensions, on the 'public attention' of a DDoS event. To quantify these dimensions, the definition of cyber harm proposed by Agraftiotis et al. [2] is mapped onto the PEST framework. The analysis chosen is the PEST analysis. The PEST analysis is a business framework that helps understand impact and relations of decisions [20]. We track these Political, Economical, Social, and Technological dimensions. These four dimensions were selected because they have previously been applied to the analysis of cyber incidents and their relationships have been explored in earlier work [5]. We use damage cues/indicators to determine severity of the damage. The cues can be seen in table 2.

We annotate the selected articles using a local LLM agent, running on the model "llama3.2:1b". The selected articles were annotated using a locally deployed LLM agent based on the Llama 3.2 1B model. This model was chosen because it can be run locally while providing adequate instruction-following capabilities for the annotation task. Each article was processed using the prompt presented in Appendix A. The LLM reads through the content of the article, which is a combination of the title and the content, and annotates the entire subset accordingly on an ordinal scale of 0-3. This scale is chosen, because on earlier tests where the scale was from 0-2, the LLM defaulted towards middle values. To make the prediction more nuanced, the scale was enlarged from 3 values, to 4 values, making it less biased towards choosing a middle values. The scale has ordinal values to make it easier for the LLM to give appropriate annotation values[23]. Each article was annotated independently to prevent the model's predictions from being influenced by previously processed articles.

3.2 Ground truth

To assess the reliability of the LLM-generated annotations, we manually annotate a random sample of 100 files based on the same cues mentioned in Table 2. Out of these 100 articles, 75 were accessible for human evaluation, due to the link being offline or behind a paywall. The manually assigned labels were treated as the reference annotations and compared with the LLM-generated labels using the Mean Absolute Error (MAE), Cohen's Weighted Kappa [6], and the mean bias for each dimension. Together, these metrics quantify the magnitude of the annotation error, the degree of agreement between the LLM and the human annotator, and the direction of any systematic differences.

4 Methodology

The objective is to extract information at the event level rather than from individual articles. Since a single article often provides only a fragmented description of an event, semantically related articles are grouped to obtain a more complete representation of the underlying DDoS incident. This event-centric view enables the impact dimensions to be analyzed more reliably. An event detection and clustering pipeline is used to semantically group articles describing the same event.

Table 2: Operationalization of PEST dimensions for cyber incident impact assessment

Dimension	Literature-based definition	Observed cyber impact characteristics
Political	National stability, geopolitical tension [8][5]. Some states even employ cyber attacks as part of broader political and strategic objectives [7].	Cyber attacks as tools in geopolitical rivalry and hybrid warfare; influence on national security agendas, international norms, and political response mechanisms.
Economical	Reputational damage, revenue loss, recovery costs, regulatory fines, and declining stock prices [2].	Macro-level modelling of extreme cyber-event losses; firm-level and sectoral financial damage; disruptions to critical infrastructure such as ports and energy systems.
Social	Societal cohesion and public distress [8].	Psychological distress, erosion of trust, weakened organisational cohesion, and societal disruption caused by attacks on critical infrastructure.
Technological	Physical and digital harm, including infrastructure disruption [2].	Impacts on availability, integrity, and confidentiality; operational inoperability of critical infrastructure; increasing socio-technical sophistication of attacks.

4.1 Event detection and tracking

The event detection and tracking component consists of a pipeline that groups semantically similar articles into clusters representing individual DDoS events. After that, we want to be able to run our regression model on the annotated data, predict annotations, and gain measurements on the performance of our predictions. A full overview of the system architecture, can be found in Figure 1.

4.1.1 Named entity recognition. It is useful for this research to recognize certain keywords within the context of an article. These keywords can give us a sense of what the article might be about; An article mentioning 'Russia' and 'Ukraine' multiple times, might be about the ongoing geopolitical conflict. To extract these entities, we used two models. We have used spaCy[16] as our general Named Entity Recognition model, and we use cyNER[3] for cybercrime related entities. SpaCy extracts ordinary entities, such as dates, countries, and numbers from the content given to it, and cyNER extracts data such as malware, systems, and organizations. Both Python libraries are industry standards, and have established use in machine learning pipelines. We store these entities in a separate cache file, where it is accessed in both the regressor and the clustering pipeline. This means that the entities are viewed two times in each pipeline, in the normal content and in the entities columns, mirroring the relevance of these keywords.

4.1.2 Semantic embedding. To identify articles describing the same event, each article is represented by a semantic embedding. Lexical similarity

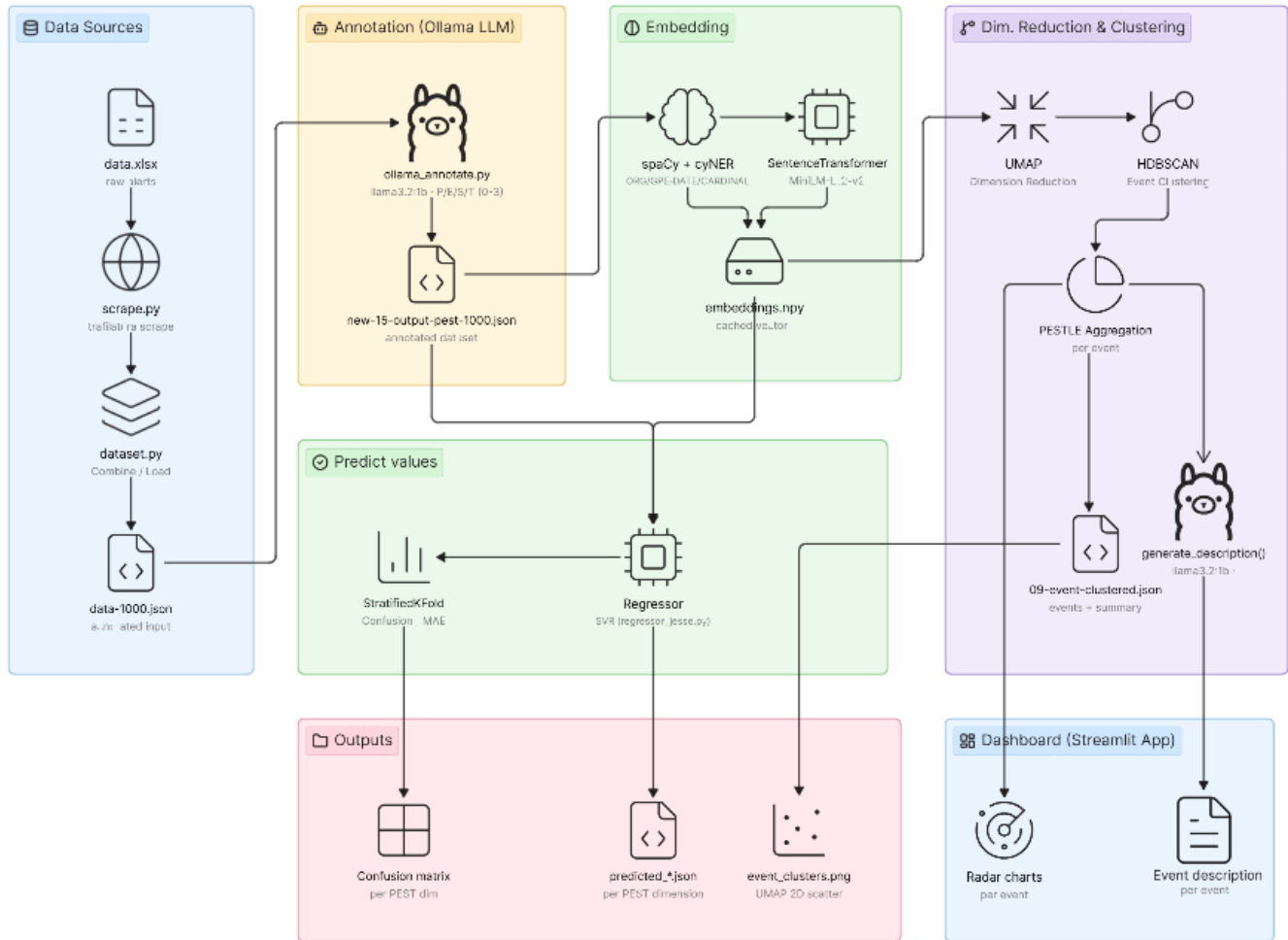


Figure 1: ML Pipeline — System Architecture

alone is insufficient, as different news sources often use different wording to describe the same DDoS incident. Semantic embeddings capture the meaning of a text rather than its exact word choice, making them more suitable for event detection. Because our database exists of raw Google Alerts, and DDoS attacks occur in all countries and thus in many languages, it is important for us to use a multilingual embedding model. We used the 'paraphrase-multilingual-MiniLM-L12-v2' model from the SentenceTransformers framework to represent textual items in the topic detection pipeline, which is a multilingual model based on the Sentence-BERT embeddings architecture from Reimers and Gurevych[18]. Each article is encoded as a high-dimensional vector, where semantically similar articles are positioned close to one another in the embedding space. This density can be mathematically measured in our clustering algorithm via cosine similarity. This metric measures the difference in the angle of a vector[19], instead of the length of the vector, making it better for high dimensional embedded vectors[13]. Articles with high cosine similarity are therefore more likely to describe the same DDoS event and can subsequently be grouped by the clustering algorithm.

4.1.3 Event clustering. Now that the content of our dataset is mapped in a vector space, we can cluster vectors in groups we call 'events'. To visualize this vector space in a 2D image, we use UMAP[15], an open software package

widely used when using our clustering technique. UMAP helps us gain a better understanding of the structure of our clusters.

Because the embeddings created are mapped on a multidimensional vector plane, we want to group these vectors based on how close they are on the vector plane. For this we use a Density based clustering algorithm. Specifically, we use HDBSCAN[14] to group articles together, because it is good in clustering high dimensional data, without knowing how many clusters exist in the database, unlike K-means clustering. HDBSCAN is a density based algorithm, which means that points become core points if they satisfy density conditions defined by the epsilon neighborhood and minimum sample parameters. The minimum number of vectors needed for a cluster, and the proximity needed to belong to a cluster, can be manually adjusted.

4.2 Machine Learning Regression

To investigate whether the semantic content of DDoS news articles is predictive of their PEST impact scores, a regression pipeline was developed and evaluated. The underlying assumption is that if article content reliably reflects PEST impact, a model trained on sentence embeddings should be able to predict annotated scores with reasonable accuracy. This section

Table 3: Distribution of annotation scores across the four dimensions.

Dimension	Score 0	Score 1	Score 2	Score 3
Political	569 (66.2%)	16 (1.9%)	223 (25.9%)	52 (6.0%)
Economical	549 (63.8%)	79 (9.2%)	162 (18.8%)	70 (8.1%)
Social	637 (74.1%)	136 (15.8%)	61 (7.1%)	26 (3.0%)
Technological	184 (21.4%)	49 (5.7%)	162 (18.8%)	465 (54.1%)

describes the Support Vector Regressor used for this purpose and the validation metrics applied to assess its performance.

4.2.1 Linear Support Vector Regression. Prior work on the same dataset, found linear kernels most effective for classification[11]. Since the present study predicts ordinal PEST scores rather than discrete classes, linear regression models were considered the most appropriate starting point. Because the sentence embeddings are represented in a 384-dimensional feature space, linear models also provide a computationally efficient and robust choice. We have two candidates: Ridge Regression and Linear SVR. Both candidates were compared via nested cross-validation, which is essential to gain the best setup[22]. Linear SVR is the regression counterpart of the Linear SVM shown to be most effective for classification on this dataset in prior work.

These type of algorithms can deduce rules based on both input and output. It requires known, correctly annotated data, which is the output it uses, to be used as labels to learn how to predict values in similar articles. Since we want to gain insight into several dimensions at the same time, it is best practice to train a separate model for each annotated/predicted dimension. In our pipeline, we implement a 'build best pipeline' function, which helps us achieve optimal parameters for our pipeline, distinguishing modeling problems from label quality problems.

4.2.2 Performance measurement. The performance of the algorithm is measured via three distinct metrics[9]. We use the Mean Absolute Error (MAE) which shows directly how many points an average prediction is off. The Rooted Mean Squared Error (RMSE) to see if the algorithm makes consistent small mistakes, or incidental large mistakes. Spearman [21] helps us see if the algorithm accurately predicts and maintains relative impact. This is important for our clustering, since the relative differences are more important in that pipeline, rather than the absolute impact. For the MAE and the RMSE it is better to have a low value, while the Spearman value is better when it gets closer to 1.

5 Results

5.1 Dataset annotation

In the dataset, the distribution of the annotated scores is shown in Table 3. Notably, the Technological dimension is inverted relative to the others, with 54.1% of articles receiving the maximum score of 3.

To verify ground truth of our LLM annotated dataset, we hold it against the subset of the dataset which has been manually annotated. The results can be seen in Table 4. The MAE represents the average absolute difference in annotation score. Cohen's weighted κ measures agreement beyond chance, where higher values indicate stronger agreement. The bias direction tells us about the direction of the LLM annotation: Is it consistent to one side, or is it balanced in its mistakes. A positive score means the LLM is biased towards picking a higher number than the ground truth annotation, a negative score means that LLM consistently picks a lower value than the ground truth. Overall, the results in Table 4 show low agreement between the LLM-generated annotations and the manually annotated reference labels, as indicated by the Cohen's values across all dimensions.

Table 4: Performance metrics and bias direction across training dimensions.

Dimension	MAE	Cohen's κ	Bias Direction
Political	1.03	0.164	-0.39
Economical	0.96	0.207	+0.29
Social	0.85	0.141	-0.59
Technological	1.43	0.057	+1.03

The Mean Absolute Error (MAE) indicates that predictions deviate by approximately one score point on average on the 0–3 scale. The bias analysis shows systematic deviations between predicted and manual annotations that differ per dimension. The Political and Social dimensions show negative bias values, indicating that the LLM tends to assign lower scores than the manual annotations. The Economical dimension shows a small positive bias, while the Technological dimension exhibits the largest positive bias among all dimensions.

5.2 Prediction power

We measure the prediction power of our algorithm using the MAE, RMSE, and the Spearman correlation. We measure each metric separate on each dimension. The results of the cross-validation are shown in Table 5. The Social dimension achieves the lowest MAE (0.592 ± 0.029) and RMSE (0.928), while the Technological dimension shows the highest error across all metrics ($MAE = 1.023 \pm 0.036$, $RMSE = 1.316$). Spearman correlation values are generally low across all dimensions, indicating weak rank-order agreement between predicted and actual scores.

Table 5: Cross-validation performance metrics across training dimensions.

Dimension	MAE	RMSE	Spearman
Political	0.852 ± 0.057	1.159	0.188
Economical	0.874 ± 0.045	1.152	0.140
Social	0.592 ± 0.029	0.928	-0.003
Technological	1.023 ± 0.036	1.302	0.140

Figure 2 presents the aggregated confusion matrix comparing predicted and actual PEST scores across all dimensions. The matrix shows that predictions are heavily concentrated around the mid-range class (score 1), while extreme values (0 and 3) are predicted less frequently. In particular, score 3 is rarely predicted except in the Technological dimension, where higher scores are more frequently assigned. Across Political, Economic, and Social dimensions, actual scores of 0 are often predicted as 1, and actual scores of 2 are also frequently mapped to 1, indicating a tendency toward central predictions. Overall, most prediction errors occur between adjacent classes, suggesting that misclassifications are generally close to the true values rather than extreme deviations.

Act 0	P	233	276	59	1
	E	193	307	45	4
	S	356	275	6	0
	T	3	44	109	28
Act 1	P	8	8	0	0
	E	29	43	6	1
	S	69	63	1	3
	T	2	6	35	6
Act 2	P	59	109	45	10
	E	32	101	26	3
	S	32	29	0	0
	T	1	18	108	35
Act 3	P	4	30	18	0
	E	7	52	9	2
	S	10	16	0	0
	T	10	59	282	114
		Pred 0	Pred 1	Pred 2	Pred 3

Figure 2: Confusion matrices per dimension

Most predictions are adjacent to the actual values. We see that Political, Economical, and Social actual 0 values, often get predicted as 1 values. Actual Political and economical value 2, also gets predicted often as a 1 value.

5.3 Event pipeline

In Figure 3, we see an overview of our UMAP reduces vector space. UMAP has brought the multidimensional sentence-embedded vector space down to a 2 dimensional plane, where we can see all the separate event clusters. The x- and y-axes are undefined dimensions, arbitrarily chosen by UMAP to provide insight into this image. These dimensions have the highest difference/information gain of all 308 dimensions in our vector space, but cannot be labeled by humans.

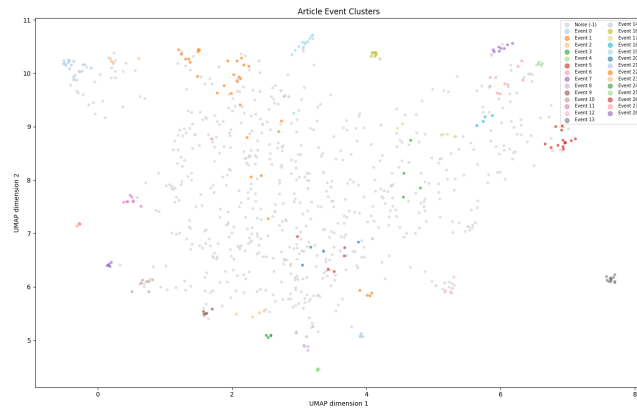


Figure 3: Event clusters

The pipeline identified 28 distinct events from 860 articles, with 652 (75.8%) articles classified as noise. Cluster sizes ranged from 4 to 28 articles (mean 7.4, median 6).

6 Conclusions

6.1 General conclusions

6.1.1 Dataset annotation. Our Large Language Model based annotation is not a reliable substitute for human judgment across dimensions. The agreement between dimensions ranges from poor to near-zero. Cohen's κ is considered poor below a value of 0.2.

Economical has the highest κ , and a low MAE value, and the bias is relatively balanced. Political is close in terms of κ and bias, but the MAE is a lot higher. In Social we can see concrete problems arising: The bias of the LLM has a substantial direction bias. The Technological dimension is the most problematic dimension. The κ is near-zero, the annotation is on average of by almost 1.5 point, and the model has a large bias of more than 1 point, meaning the LLM treats most articles as substantially more technical than the ground truth would. Overall, the results indicate that the LLM consistently overestimated the technological relevance of news articles compared with the human annotations.

6.1.2 Prediction power. The Support Vector Regressor showed limited ability to predict PEST dimension scores from sentence embeddings. Across all five stratified cross-validation folds, predictive performance was weak, with all four dimensions exhibiting weak to negligible Spearman rank correlations between predicted and annotated scores. The Political ($\rho = 0.188$, MAE = 0.852) and Economical ($\rho = 0.140$, MAE = 0.874) dimensions achieved the highest correlations, although predictive performance remained poor, with average errors exceeding 0.85 points on a 0–3 scale. The Social ($\rho = -0.003$, MAE = 0.592) and Technological ($\rho = 0.140$, MAE = 1.023) dimensions showed little to no predictive relationship between the sentence embeddings and the annotated scores.

6.1.3 Article to event pipeline. It is possible to create event level insights out of an article level dataset, by employing an event recognition pipeline, using semantic embedding and density based clustering. The pipeline classified 28 separate events in a dataset of 860 relevant articles, with 652 (75.8%) noise articles, indicating that some DDoS related coverage clusters around distinct events, but that most relates to general reporting. The events found are tight clusters, which confirms that entity-enriched embeddings have enough signal-strength to support event-level analysis, without manual article labeling. The remaining noise fraction however, is substantial.

6.1.4 Artifact description. The developed artifact, which is showcased in depth in section B, will be made public¹ after publication of this article. The artifact demonstrates that DDoS-related datasets can be transformed into structured event-level insights through an automated pipeline combining LLM-based annotation, semantic embeddings, and density-based clustering. The resulting system supports not only event detection, but also higher-level interpretation through visualization and PEST-based impact analysis. This shows the potential of the approach to reduce manual labeling effort while enabling scalable analysis of large-scale cybersecurity datasets. To deploy the artifact, the user needs a Python environment to run the python files, and an Ollama account for its annotation, for which a stable internet connection is needed. The verification of 5 out of 28 events can be found under section C.

6.2 Discussion

6.2.1 Dataset annotation. In comparable fields, LLM annotation reportedly had κ values of 0.56–0.75 [17][10], which is substantially above our values of 0.057–0.207. However, these studies involved binary or categorical labeling tasks, where our annotation requires ordinal labeling. This distinction likely explains the gap, since ordinal values provide more discussion on annotation tasks, even among human annotators [4]. If humans have difficulty consistently annotating these types of tasks, a small LLM model like 'llama3.2:1b' experiencing problems is expected.

6.2.2 Prediction power. The relatively low MAE for Social is likely explained by the model defaulting toward the majority class rather than learning true variation in scores, a pattern consistent with imbalanced label distributions. The Technological dimension produced both the highest

¹https://github.com/jesuisvandijk/DDoS_Impact_Visualizer

MAE (1.023) and the largest RMSE (1.302), indicating not only poor average accuracy but frequent large prediction errors. Taken together, these results suggest that PEST dimensions are not reliably predictable from sentence embeddings alone at the current sample size and class distribution. No dimension demonstrates even moderate predictive validity, which points to a need for either richer feature representations, a larger and more balanced annotated dataset, or an alternative modeling approach.

It should be noted that the regression serves primarily as a validation tool rather than a production component. Poor predictive performance would not necessarily indicate a flaw in the embedding approach itself, but could equally reflect inconsistencies in the training labels produced by the LLM annotation pipeline. Automated annotation at scale introduces noise, and if the LLM assigned scores inconsistently across similar articles, no downstream model can be expected to learn a stable signal from that data. The results of this validation step should therefore be interpreted with that caveat in mind. This conclusion is supported by the 'build best pipeline' function from the pipeline. In here, several parameters are tested, but it makes virtually no difference. This suggests that the configuration was already near optimal, meaning performance limitations stem from label quality rather than modeling choices.

6.2.3 Article to event pipeline. HDBSCAN's documentation explains that 40% of noise indicates parameter misconfiguration or significant lacking data structure. Our 75.8% is well above that threshold. This might reflect isolated stories, unrelated articles, or HDBSCAN's `min_cluster_size` threshold excluding small but legitimate clusters. The `min_cluster_size = 4` was a deliberate, conservative choice. Relaxing this setting would introduce noise into cluster coherence. We preferred false negatives, above false positives, to increase the usability of the artifact.

6.3 Limitations

Several limitations have been found in this research. Due to limited processing power, for several used models lighter variants were selected due to constraints. This case is true for the LLM model, the SentenceTransformer model, and the cyNER model. All of these models have been tested at an expanded model, but eventually were used on the base model. Processing power would have increased the model power, making it probably more accurate. Reliably gaining verifiable annotation by the chosen LLM proved to be rigorous. Balancing the instructions of the prompt with the context of the article and the assignment tasked to the LLM, all while maintaining token limit and speed, was a long process in this research. Finally, the use of a discrete impact scale introduces a degree of subjectivity in the evaluation of results. This limitation could be reduced through a larger number of manually annotated cases, which would likely improve the robustness and reliability of the impact assessment.

6.4 Future work

In this research, only one regression model has been used and tested. In future research, comparing several advanced regression models against each other, and choosing the strongest model, would make predictions on other datasets more accurate, and more usable. It might also be interesting to look at the power of the enrichment done by the entities. Increasing the number of times the entities are injected into the content, might severely influence the performance of the model. Lastly, the model has been trained on a Google Alerts dataset concerning DDoS events. While this provides a scalable and accessible source of data, it may also introduce noise and broader contextual variation. Future research could therefore benefit from using more strictly filtered datasets that focus exclusively on confirmed DDoS attack events.

7 Acknowledgments

We thank Abhishta for his guidance during this research. His support has led to many insights. We thank Jan-Willem for his feedback, which led to an improved version of this article.

I would like to thank my housemates and my parents for their continuous support and patience. I thank Jasmijn especially for her revisions and suggestions. The research would not be the same without you. During the preparation of this work the author used Anthropic's Claude Sonnet 4.6 in order to interpret large Python repositories, Ollama's Llama3.2:1b in order to annotate the dataset, and generate descriptions of each event, and ChatGPT to rewrite the references table in proper $\LaTeX$ format. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the work.

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A Prompt

The following text is the exact prompt instructing the local AI agent to annotate each article:

“”””You are an expert analyst specializing in cybersecurity impact assessment. You will read a news article (title + body) and independently score it on four dimensions.

— SCORING SCALE:

- 0 = No meaningful content relating to this dimension
- 1 = Minor or indirect mention (topic is peripheral)
- 2 = Moderate coverage (topic is discussed but not the main focus)
- 3 = Primary focus (topic is central to the article)

Treat each dimension as INDEPENDENT. A score on one must not influence another. Use the full range 0–3. Do not default to middle values.

— DIMENSIONS:

[POLITICAL]

Damage: national stability, geopolitical tension, political strategy regarding cyber attacks. Indicators: cyber attacks as tools in geopolitical rivalry and hybrid war; influence on national security agendas and international norms; political reaction.

- 0 – Article covers a technical vulnerability with no political actors or reactions mentioned.
- 1 – A government agency is named as affected, but political implications are not discussed.
- 2 – Article discusses a nation-state attribution or calls for a policy response.
- 3 – Article centers on geopolitical conflict, diplomatic fallout, or national security strategy driven by a cyber event.

ECONOMIC

Damage: reputational damage, revenue lost, recovery costs, fines, stock price lowers. Indicators: macro loss modelling of extreme cyber events; firm level and sectoral losses; critical infrastructure disruptions (e.g., ports, energy).

- 0 – No financial figures, losses, or economic actors mentioned.
- 1 – A company is named as victim but financial impact is not quantified or discussed.
- 2 – Article reports estimated losses, fines, or disruption to a specific firm or sector.
- 3 – Article focuses on large-scale financial damage, systemic economic risk, or critical infrastructure with major economic consequences.

SOCIAL

Damage: societal cohesion, public distress. Indicators: psychological distress, loss of trust, weakened organisational cohesion, societal disruption from critical infrastructure attacks.

- 0 – No mention of public impact, trust, or civilian disruption.
- 1 – Brief mention of public concern or inconvenience without elaboration.

- 2 – Article discusses erosion of trust or distress in a specific community or organisation.
- 3 – Article centers on widespread societal disruption, mass loss of public trust, or psychological impact at scale.

TECHNOLOGICAL

Damage: physical/digital harm, infrastructure disruption. Indicators: availability, integrity, and confidentiality impacts; critical infrastructure inoperability; socio-technical sophistication.

- 0 – No technical detail; article is purely political or economic commentary with no system impact.
- 1 – Attack type is named (e.g. "ransomware") but technical details are absent.
- 2 – Article describes attack vectors, affected systems, or partial infrastructure disruption.
- 3 – Article provides in-depth technical analysis, covers novel TTPs, or reports full infrastructure inoperability.

— INSTRUCTIONS:

- Base your scores on the article content only. Do not infer beyond what is written.
- Weight sustained, substantive coverage more than passing mentions.
- If the title contradicts the body, follow the body.
- Score each dimension independently before combining into the final JSON.

Respond ONLY with a valid JSON object. No explanation, no markdown, no extra text:

"Political": <0-3>, "Economic": <0-3>, "Social": <0-3>, "Technological": <0-3>""

B Artifact showcase

B.1 Using artifact on own dataset

To use the artifact for personal use, the following steps should be followed: Upload the Google Alert data into the folder of the artifact called 'Data'.

The file can be named whatever is desired, but should use '.xlsx' as an extension. We will use 'data.xlsx' as an example file. The data should consist of normal Google Alert output columns. The artifact needs at least the title, content, link, data, and type column.

The desired data can be scraped using the 'scrape.py' file. Scrape.py scrapes the content from the articles, using the link, and inputs it back into the 'content' column, enriching the 'content' column with more information.

The file can be used via the terminal in the preferred python environment. The command looks something like this:

```
python scrape.py -input data.xlsx -output new_dataset.json -type-filter News -max-rows
```

The 'input' file is the data file uploaded into the artifact folder. The 'output' file is the file used later into the pipeline, and will be created upon running the statement. '-type-filter' decides whether the artifact only scrapes news articles, or blogs or other articles will also be used. The '-max-rows' statement decides how many rows will be scraped into the new JSON file. Next, the 'ollama_annotate.py' should be ran. First, update the input_path= to look like the output file chosen in 'scrape.py'. The output_path= has to be edited in the 'article_to_event_level.py', and it should be a new file, which will be created in the 'Data' folder of the artifact. The filename can be chosen as desired. This file annotates the dataset with values for each PEST dimension between 0 and 3. This takes a very long time, depending on the size of the dataset.

Lastly, the 'article_to_event_level.py' can be ran. This file has several settings which can be chosen for the specific dataset. These settings decide the closeness and dimensionality of the dataset. Edit when the results of the pipeline look wrong. There are also toggleable settings, choosing to

generate descriptions for each event, to launch the dashboard, and to show the graphs created in the python environment. Generating descriptions can be very slow, but they are stored in the cache file. These cache files should be deleted when using a new dataset.

B.2 App showcase

In this section we look through the artifact created as in this research. We will go through the dashboard, and explain the artifacts usability. The artifact is loaded at the end after running the `article_to_event_level.py` file, or after running the `streamlit_app.py` file. The app should open a browser-tab, and display the following homepage:

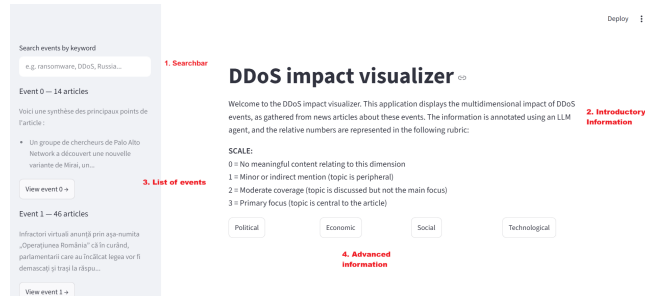


Figure 4: Landing page of the artifact

On this page, several relevant items have been highlighted:

- (1) The searchbar is used to search for keywords in the descriptions of events. It searches for semantically similar keywords, using sentence embedding. It is accessed in the sidebar, which can be toggled, and can be made visible from each event-page.
- (2) The landing page includes a short introduction to the usability of the page, as well as basic information about what each score means.
- (3) A list of events is displayed. It is accessed in the sidebar, which can be toggled, and can be made visible from each event-page. Each event has a short description and a button, to directly access information about this event.
- (4) Additional information about separate dimensional scoring rubrics can be accessed

After clicking on an event in the sidebar, the user is taken to a new screen. This screen shows more detailed information about the event.

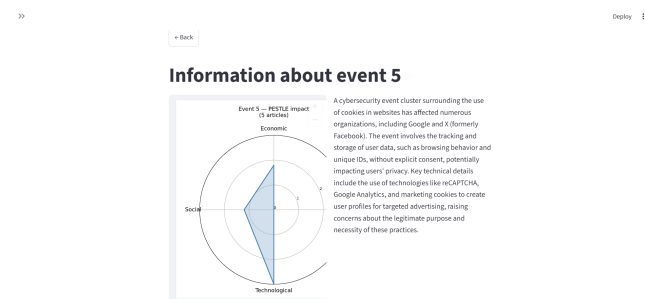


Figure 5: Event overview

In Figure 5, we see a short overview of the most important aspects of an event. On the left, a radarplot has been created from the average PEST dimensions of the articles included in this event, and on the right there has been generated a description about the event. This description is based on

the full content of all the articles in this event, and has been generated by the Large Language Model 'llama3.2:1b'. The prompt is as follows: "You are given (*number of articles*) news articles that belong to the same DDoS event cluster. Summarize this DDoS event in 3 sentences in English: what happened, who was affected, key technical details. Start directly with the description, no introduction, presenting or affirmation." Below the legend of this event, a table is shown where additional information can be displayed.

The 'Event table' showing a list of articles. The table has columns for Content, Date, and Source. The articles are listed in descending order of their impact score. The first article is 'To provide the best experiences, we use technologies like cookies to store and/or access information on your device to enhance navigation and site usage, and to improve our marketing efforts. For more information, please see our privacy policy.' The second article is 'Websites store cookies to enhance functionality and personalize your experience. You can manage your preferences at any time. For more information, please see our privacy policy.' The third article is 'To provide the best experiences, we use technologies like cookies to store and/or access information on your device to enhance navigation and site usage, and to improve our marketing efforts. For more information, please see our privacy policy.' The fourth article is 'To provide the best experiences, we use technologies like cookies to store and/or access information on your device to enhance navigation and site usage, and to improve our marketing efforts. For more information, please see our privacy policy.' The fifth article is 'May 27, 2024: Articles Blackberry's Defining What Secure Communications Actually Mean'.

Figure 6: Event table

In this table, all tabs are able to be toggled, to make sure the artifact is easy to use.

C Event verification

In here, we look at a few events to determine whether these events have rightfully been labeled a distinct event. We will look at 5 random events chosen from the list of events. We randomly select event numbers 5, 12, 14, 20, and 26.

C.1 Event 5

Event 5 is an event which contains 5 articles. In these articles, 5 separate sources report of a DDoS attack on the 'MadLads NFT mint'. On the day the NFT should have been launched, the co-owners reported unusual high internet traffic on the site, postponing the launch by an hour. The articles were uploaded between the 22nd and the 25th of April in 2023. The links to the articles are:

- <https://decrypt.co/137730/honeypot-bitch-how-mad-lads-tricked-bots-spending-250k-fake-nfts/>
- <https://www.nftgators.com/madlads-nft-public-mint-delayed-after-triggering-a-ddos-attack/>
- <https://www.coindesk.com/web3/2023/04/21/heavy-demand-for-madlads-nft-breaks-internet-delays-mint/>
- <https://cryptotoday.co.uk/2023/04/solana-based-madlads-nft-mint-pushes-through-after-delay/>
- <https://en.cryptonomist.ch/2023/04/21/big-problems-nft-market/>

C.2 Event 12

Event 12 consists of 8 articles, out of 7 separate sources. The sources report on a number of attacks on the Israeli Government, done by the hacker group 'Anonymous Sudan'. They targeted the president's Facebook and official website, an Israeli cyber-security company, and universities. The articles were uploaded between 5 and 25 April 2023. The links are:

- <https://www.secnews.gr/462153/anonymous-sudan-eptihethikan-sta-sites-tou-aerodromiou-ben-gurion-kai-tis-israel-electric-corporation/>

- <https://www.timesofisrael.com/websites-of-israeli-port-hacked-sudanese-group-said-to-claim-responsibility/>
- <https://www.i24news.tv/en/news/israel/defense/1682538792-netanyahu-s-facebook-hacked-as-part-of-independence-day-cyberattacks>
- <https://www.israelnationalnews.com/news/370588>
- <https://www.darkreading.com/attacks-breaches/anonymous-sudan-claims-responsibility-ddos-attacks-israel>
- <https://www.vesty.co.il/main/article/b15bp111m3>
- <https://www.jpost.com/breaking-news/article-736351>
- <https://www.israelnationalnews.com/news/369629>

C.3 Event 14

Event 14 spans 4 articles from 4 different sources. It depicts a 'DNS NXDOMAIN' attack on American healthcare websites. The articles were uploaded on 10 and 11 April 2023. The links are:

- <https://www.scmagazine.com/brief/threat-intelligence/us-healthcare-sector-facing-dns-nxdomain-flood-ddos-attacks>
- <https://www.healthcarefinancenews.com/news/healthcare-websites-are-being-attacked-fake-requests>
- <https://www.hipaajournal.com/hc3-warns-of-dns-nxdomain-ddos-attacks-on-the-healthcare-sector/>
- <https://healthitsecurity.com/news/dns-nxdomain-flood-ddos-attacks-impacting-healthcare-hc3-warns>

C.4 Event 20

Event 20 consists of 4 articles from 4 separate sources, published in Portuguese, Italian, and Swedish. Unlike most events in the dataset, these articles do not describe the same DDoS incident. Two articles provide general information on DDoS attacks and threat trends, while the other two report on separate incidents: a DNS flood attack against the Italian region of Friuli-Venezia Giulia and a broader DDoS campaign targeting Swedish organizations. The articles were published between 20 and 30 April 2023. The links are:

- <https://olhardigital.com.br/2023/04/20/seguranca/brasil-e-o-principal-alvo-de-ataques-ciberneticos-na-america-latina-2/>
- <https://www.difesaesicurezza.com/difesa-e-sicurezza/chitarone-2-0-cosa-sono-gli-attacchi-ddos-e-perche-non-si-possono-evitare/>
- <https://www.punto-informatico.it/friuli-venezia-giulia-attacco-sventato-alla-regione/>
- <https://www.varldenidag.se/nyheter/expert-om-attackerna-mot-svenska-sajter-en-varning/repwbt!jzehEwgMXULdp2vYxbqYg/>

C.5 Event 26

Event 26 consists of 14 articles from 14 separate sources. The sources report in multiple languages, including Spanish, English, French, German, Turkish, and Arabic. The articles describe the same DDoS attack against the European Parliament's website, which occurred shortly after Members of the European Parliament voted to designate Russia as a state sponsor of terrorism. The pro-Russian hacktivist group Killnet claimed responsibility for the attack, which temporarily disrupted access to the Parliament's website. The articles were published between 23 and 25 November 2022. The links are:

- <https://www.elfinanciero.com.mx/mundo/2022/11/24/hackean-rusos-el-parlamento-europeo/>
- <https://www.telemundowashingtondc.com/noticias/mundo/realizan-ciberataque-proruso-sitio-web-parlamento-europeo/2164866/>
- <https://www.eluniverso.com/noticias/internacional/la-pagina-web-del-parlamento-europeo-fue-victima-de-un-ciberataque-tras-voto-sobre-rusia-nota/>

- <https://www.digit.fyi/european-parliament-cyber-attack-killnet-group/>
- <https://www.whatsnew.com/2022/11/24/sitio-web-del-parlamento-europeo-fue-victima-de-un-ataque-ddos/>
- <https://www.channelnews.fr/une-attaque-ddos-contre-le-parlement-europeen-119882>
- <https://thenextweb.com/news/russian-cyberattack-eu-parliament-shows-need-for-better-public-sector-security>
- <https://www.computing.co.uk/news/4060707/european-parliament-website-suffers-sophisticated-cyber-attack-russia-terrorism-vote>
- <https://www.ndtv.com/world-news/eu-parliament-website-cyberattack-claimed-by-pro-russian-hackers-3547964>
- <https://www.politico.eu/article/cyber-attack-european-parliament-website-after-russian-terrorism/>
- <https://www.trtworld.com/europe/pro-kremlin-hackers-target-eu-parliament-website-after-vote-against-russia-62826>
- <https://www.republicworld.com/world-news/russia-ukraine-crisis/european-parliament-website-affected-by-hacking-attack-by-pro-moscow-group-articleshow.html>
- <https://www.dw.com/en/european-parliament-hit-by-sophisticated-cyber-attack/a-63864029>
- <https://saudigazette.com.sa/article/627276/World/Europe/EU-Parliament-website-hacked-after-MEPs-passed-critical-Russian-resolution>